

Multi-choice questions

1. What does the `exec()` system call do when it succeeds?
- a) Returns a value indicating success
 - b) Replaces the current process image and never returns
 - c) Creates a child process
 - d) Pauses the current process

Answer: b

2. Which statement best describes what happens when a process calls `exit()`?
- a) The process is paused until its parent process collects the status value
 - b) The process terminates normally, performs process cleanup, and makes the exit status available to the parent process
 - c) The process terminates only if all child processes have already completed
 - d) The process replaces its current memory image with another program while preserving the same PID

Answer: b

3. Which statement correctly includes the standard C++ I/O header?
- a) `#import <iostream>`
 - b) `#include <iostream>`
 - c) `import iostream;`
 - d) `include iostream;`

Answer: b

4. In the following code segment, what does the operator `>>` do?

```
int x;  
std::cin >> x;
```

- a) Inserts data into the `cin` stream
- b) Extracts data from the `cin` input stream into `x`
- c) Performs a bitwise shift operation on `x`
- d) Creates a new input/output stream

Answer: b

5. Which statement best describes the relationship between C and C++?
- a) C++ is a strict superset of C
 - b) Every valid C program is also a valid C++ program
 - c) Many C programs can compile in C++, but some C programs are not valid C++
 - d) C++ only supports object-oriented programming

Answer: c

6. What is the primary purpose of namespaces in C++?
- a) To control the lifetime and visibility of variables within functions
 - b) To group related identifiers and reduce the likelihood of name conflicts across a program
 - c) To provide access control for class members and inherited classes
 - d) To organize input/output stream objects used by the standard library

Answer: b

7. What is printed by the following code?

```
#include <iostream>
namespace A { int value = 10; }
namespace B { int value = 20; }
int main() {
    std::cout << B::value;
    return 0;
}
```

- a) 10
- b) 20
- c) 30
- d) Compilation error

Answer: b

8. What happens when the following C++ program is compiled?

```
#include <iostream>
using namespace std;

namespace A { int value = 5; }
namespace B { int value = 10; }

using namespace A;
using namespace B;

int main() {
    cout << value;
}
```

- a) The program prints 5
- b) The program prints 10
- c) The program prints 510
- d) The program results in a compilation error

Answer: d

9. In a C++ class, what is the default access level if no access specifier is written?

- a) public
- b) private
- c) protected
- d) static

Answer: b

10. Which standard C++ stream is typically used for normal output to the console?

- a) cin
- b) cout
- c) cerr
- d) clog

Answer: b

11. How many times is the default constructor of class `Time` invoked in the following program?

```
class Time {  
public:  
    Time() {}  
};  
  
int main() {  
    Time t1;  
    Time t2;  
}  
a) 0  
b) 1  
c) 2  
d) 3
```

Answer: c

12. Which constructor of class `Student` is invoked when the object `obj` is created in the following program?

```
class Student {  
public:  
    Student() {}  
    Student(int id) {}  
};  
  
int main() {  
    Student obj(10);  
}  
  
a) Default constructor  
b) Parameterized constructor  
c) Copy constructor  
d) Destructor
```

Answer: b

13. What does the `const` qualifier at the end of the following member function declaration indicate?

```
class Time {  
public:  
    void print() const;  
};  
  
a) The function must return a constant value  
b) Objects of the class cannot be modified  
c) The member function does not modify the object's member variables  
d) The member function becomes static
```

Answer: c

14. Which statement about inline functions in C++ is most accurate?

- a) A compiler must always honour an inline request
- b) Inline functions are guaranteed to execute faster than regular functions
- c) Defining a member function inside a class definition is an implicit request to inline the function
- d) Member functions cannot be declared as inline

Answer: c

15. Which standard C++ stream is typically used for reading input from the keyboard?

- a) cin
- b) cout
- c) cerr
- d) clog

Answer: a

16. Which statement best describes memory management in C++ compared with Java?

- a) C++ and Java both primarily rely on automatic garbage collection
- b) C++ generally provides programmers with more explicit control over memory through pointers and manual resource management
- c) C++ does not support dynamic memory allocation
- d) Java requires programmers to explicitly free dynamically allocated memory

Answer: b

17. Conventionally, what does an exit status value of 0 indicate when a process terminates?

- a) The process terminated due to an error
- b) The process terminated successfully
- c) The process was terminated by a signal
- d) The child process is still executing

Answer: b

18. State True or False. Friend functions in C++ can access private and protected members of a class

Answer: True

19. State True or False. In C++, objects are created on the heap by default when declared as regular variables.

Answer: False

Short answer questions

20. In a parent-child process relationship, what does `wait()` allow the parent to do?

Answer: Suspend execution until a child process terminates and collect its exit status.

21. Give two reasons why a compiler might choose not to inline a function.

Answer: 1) The function is recursive

2) The function contains loops

3) The function contains static variables

22. Why does the following code segment result in a compilation error?

```
class Test {  
    public:  
        void print() const {  
            x = 5;  
        }  
  
    private:  
        int x;  
};
```

Answer: The member function is declared `const`, so it is not allowed to modify member variables such as `x`.